Formal Methods: Logic as the Foundation

Chapter 5 (5.1–5.7) – Extended Lecture Notes



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Logic evolved over two millennia

01 **Aristotle**

Syllogistic reasoning with categorical statements

02 **Stoics**

Propositional logic and conditional statements

03 **Boole**

Algebraic formalization with 0 and 1

04 **Frege**

First-order predicate logic with quantifiers

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Aristotle's syllogisms: The birth of deduction

01 Syllogism Structure

Two premises leading to a conclusion through deductive reasoning

02 Classic Form

All A are B; All B are C; Therefore All A are C

03 Categorical Statements

All, Some, None – forms the basis of logical classification

04 Foundation of Deduction

Foundation of deductive reasoning for centuries

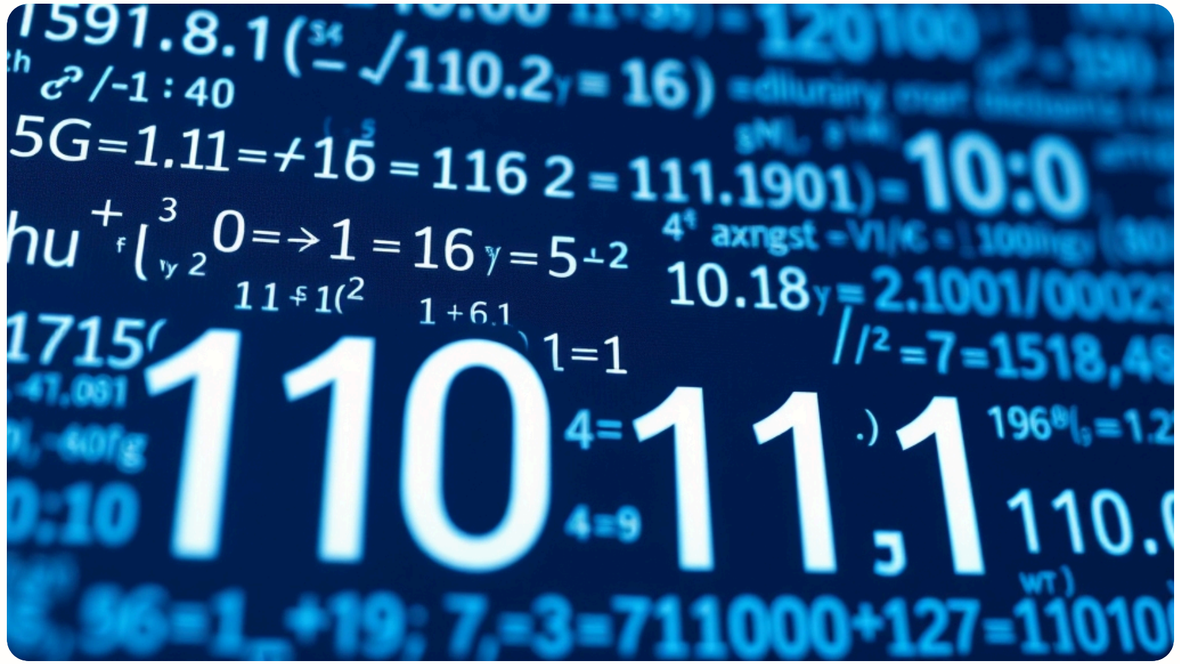
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From Stoics to Boole: Logic becomes mathematical



Stoic Logic

Conditional reasoning (If p then q). Early Modus Ponens and Modus Tollens introduced. Focused on propositions and their relationships rather than categorical statements.



Boolean Algebra

Logic expressed with 0 and 1. Foundation for digital circuit design. Transformed logic from philosophy into mathematics, enabling computational applications.

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Propositional Logic: Building blocks of reasoning



Propositions

Statements with True or False values



Connectives

NOT, AND, OR, IMPLIES, IFF combine propositions



Truth Tables

Define semantics and evaluate complex expressions



Classifications

Tautology, Contradiction, Contingency categorize formulas

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Logical Equivalence simplifies expressions

01

Equivalent Statements

Equivalent statements share identical truth tables

02

De Morgan's Laws

¬(p ∧ q) ≡ ¬p ∨ ¬q

03

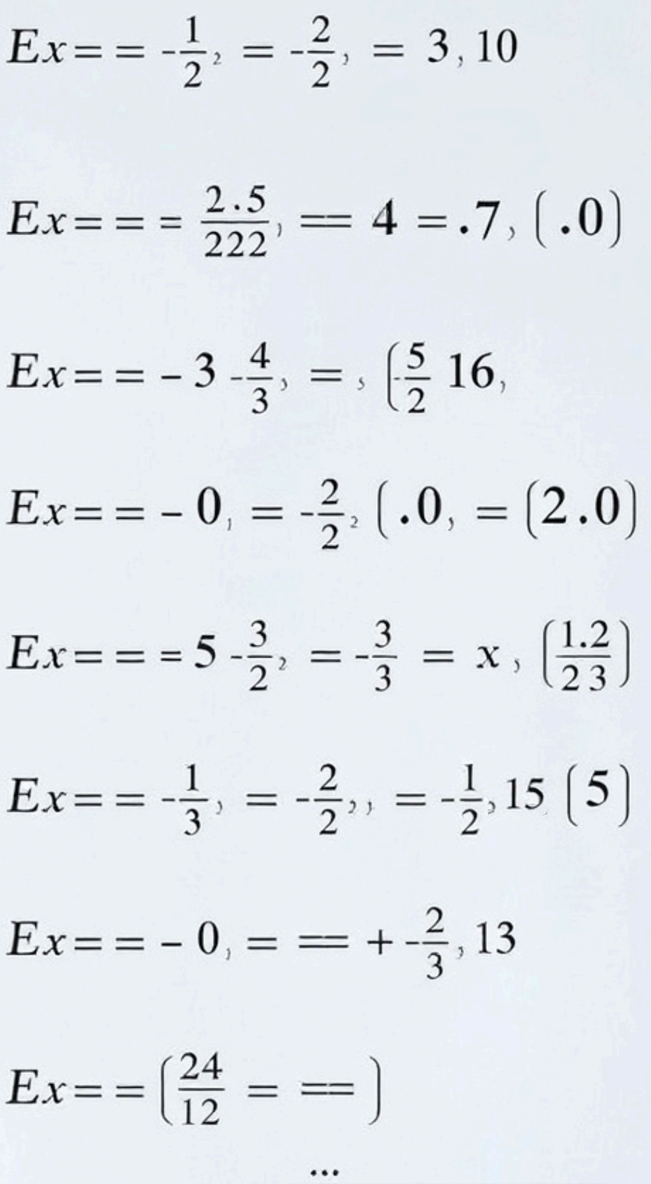
Implication Equivalence

p → q ≡ ¬p ∨ q

04

Practical Use

Used to simplify and transform logical expressions



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Predicate Logic: Expressive power with quantifiers

Variables and Predicates

Extends propositional logic with variables and predicates

Universal Quantifier

For all (∀)

Existential Quantifier

There exists (∃)

Domain of discourse defines variable scope. Negating quantifiers: Flip quantifier and negate predicate.



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Inference and Proof Techniques

Modus Ponens

If p→q and p, then q

Modus Tollens

If p→q and ¬q, then ¬p

Direct Proof

Assume premises, derive conclusion

Proof by Contradiction

Assume negation, find contradiction. Also: Proof by Contrapositive: Prove ¬q → ¬p

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Applications

Logic powers formal specification

→ **Model Checking**

Model checking verifies system properties

→ **Safety-Critical Systems**

Safety-critical systems require mathematical correctness

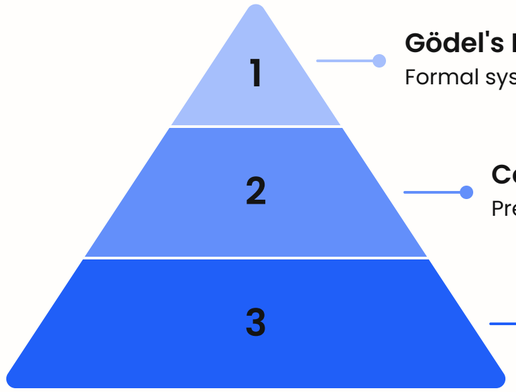
→ **Specification Languages**

Specification languages built on logical foundations. Ensures software behaves as intended.

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Limitations: What logic cannot solve

Understanding the boundaries of formal systems



Gödel's Incompleteness
Formal systems have inherent limits

Computational Limitations
Predicate logic has computational limitations. Some mathematical truths are undecidable.

Expressive Limits
Propositional logic lacks expressive power for complex statements

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